



- Merchant signup with KYC fields (business name, type, GST ID, etc)
- Login with email + password (BCrypt encoded).
- API key generation with show-once secret.

Order Lifecycle

- Create order with amount, currency, receipt, notes, expiry.
- List orders with pagination and filtering.
- Get order by ID.
- Order auto-expiry after 15 minutes.
- Idempotent order creation via X-Idempotency-Key header.

Payment Lifecycle

- Initiate payment for an order with method + method details.
- Support for Card, UPI Intent, Net Banking payment methods.
- Mock acquirer with chaos modes (SUCCESS, FAILURE, TIMEOUT, SLOW, NORMAL).
- Payment state machine with 8 states + valid transitions enforced.
- Idempotent payment APIs via X-Idempotency-Key.

Refund Lifecycle

- Full and partial refunds.
- Refund sub-state machine (PENDING → PROCESSING → PROCESSED / FAILED).
- Async refund processing via scheduled job.

Webhook Delivery

- Per-merchant webhook URL configuration.
- HMAC-SHA256 signing of payloads.
- 7-attempt retry with exponential backoff (immediate, 1m, 5m, 30m, 2h, 8h, 24h).
- Dead Letter Queue after 7 failures.
- Replay API for dead-lettered webhooks.

Settlement

- Nightly batch settlement at 11pm (configurable).
- Fee calculation (2% of captures).
- GST calculation (18% of fee).
- Per-merchant net settlement with per-merchant failure isolation.
- Settlement record with full audit trail.
- Mock bank transfer with chaos.

Card Vault

- Tokenization API (browser → vault direct, never via merchant)
- AES-256 encryption of PAN.
- Charge-with-token operation (no get-PAN API exists).
- Customer-saved tokens for one-click checkout.
- @MaskedCard annotation + Logback filter for log safety.

Analytics

- Real-time dashboard with revenue today.
- Per-merchant analytics filter.
- Historical reports (last 30 days).

Multi-Tenant Security

- API key auth (Basic auth header) for server-server endpoints.
- JWT auth for dashboard endpoints.
- Per-merchant rate limiting (token bucket, sliding window, fixed window).

